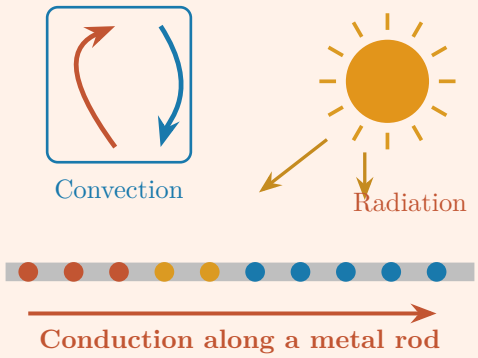
NCERT CURIOSITY • CHAPTER 7

CLASS 7 SCIENCE

HEAT TRANSFER IN NATURE

Conduction • Convection • Radiation
Land and sea breeze • Water cycle



How this booklet is organised

First: every numbered NCERT exercise question (with reasons and picture-based answers).
Then: concise teaching notes, colourful diagrams, common errors, memory tricks and a revision check.

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Textbook exercise answers (pp. 102–104)

Q1 (i)–(iii) • Choose the correct options

(i) Saucepan with parts A and B (Fig. 7.14).
Answer. (c): A (the pan body) is a **good conductor**; B (the handle) is a **poor conductor**.
Why? The pan must pass heat to the food, while the handle should reduce heat flow to the hand.
(ii) Wax-attached pins on a heated metal strip (Fig. 7.15).
Answer. (b): pins I and II fall earlier than III and IV.
Why? Heat conducts from the candle end towards the other end; the nearer wax melts earlier.
(iii) Best position for a smoke detector.
Answer. (c) On the ceiling.
Why? Warm smoke-containing air tends to rise by convection. (Follow the device manufacturer's installation instructions in real life.)

Q2 • Can a second tumbler keep leaky lassi cold?

Answer. Yes, it can help, especially if an insulating air gap remains between the two tumblers.
Why? Trapped air is a poor heat conductor, so less heat enters the cold lassi from the warmer surroundings. The effect depends on how the tumblers fit and whether liquid fills the gap.

Q3 (i)–(iv) • True or false, with corrections

Part	Result	Correct explanation
(i)	**False**	In solids, heat travels mainly by **conduction**, not convection.
(ii)	**True**	Convection moves heat by the actual movement of fluid particles.
(iii)	**False**	Water generally seeps through sandy material more readily than closely packed clay.
(iv)	**True**	A land breeze is cooler air flowing from land towards sea (often at night).

Q4 • Where does melting ice get its heat?

Answer. From the **warmer dish and surrounding air**; it also receives thermal radiation from warmer surroundings. Heat passes to the colder ice and melts it.

Q5 • Smoke from a downward-pointing incense stick

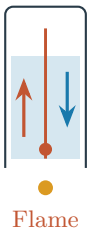
Answer. The smoke generally **rises upwards**, even though the burning tip points down. It is carried by warmer air.



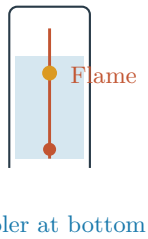
Q6 • Which thermometer becomes warmer? (Fig. 7.16)

Answer. **Figure 7.16(a)** records the higher temperature.
Why? In (a), water is heated at the **lower end** and hotter, lighter water rises to the thermometer by convection. In (b), the flame heats water **near the upper end**; the bottom thermometer is less quickly warmed because hot water stays above cooler water.

(a) Lower-end heating



(b) Upper-end heating



Q7 • Why hollow bricks in hot regions?

Answer. Their cavities hold air, which is a poor conductor. The trapped air helps **reduce heat transfer** through outer walls, keeping indoors more comfortable in hot weather.

Q8 • How do large water bodies moderate temperature?

Answer. Water warms and cools **more slowly than land**. During the day, sea breezes can bring cooler air towards shore; at night the land cools faster and the breeze can reverse. Nearby temperatures are therefore less extreme.

Q9 • How is groundwater formed?

Answer. Rainwater seeps into soil and permeable rocks through **infiltration**, moves through connected pores and openings, and collects underground in layers called **aquifers**. Wells and borewells reach groundwater stored there.

Q10 • How does the water cycle redistribute and replenish water?

Answer. Solar heating causes **evaporation** and plants add vapour through **transpiration**. Vapour cools and **condenses** into clouds; **precipitation** brings water to different places. Water returns by **runoff** and **infiltration**, refilling water bodies and groundwater.

Core ideas you need to know

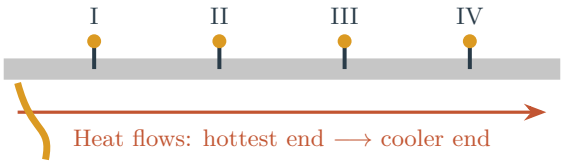
Three mechanisms of heat transfer

Method	How it works	Where and example
Conduction	Heat passes between neighbouring particles without overall movement of material.	Mainly solids: metal spoon in hot tea.
Convection	Warm fluid rises; cooler fluid sinks, creating circulation.	Liquids and gases: water in a pot, sea breeze.
Radiation	Energy travels as electromagnetic waves; no material medium required.	Sun to Earth; heat felt near a fire.

Thermal energy normally moves from hotter to colder places

A metal pan is useful because it **conducts** heat well. A wooden handle, wool and trapped air help **slow down** heat loss or gain. Heat transfer continues until temperature differences decrease.

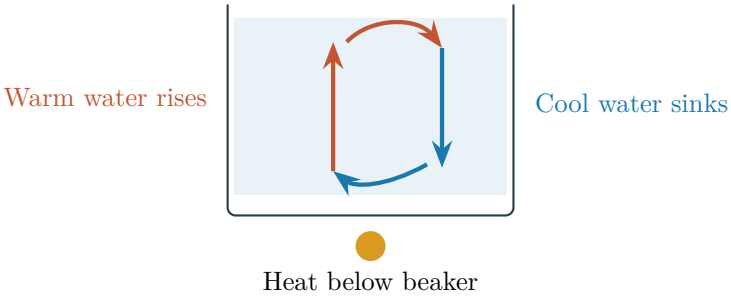
Conduction: wax and pins activity (Activity 7.1)



Predict and explain

The wax holding **pin I** melts first, then II, III and IV. The metal transfers heat successively. In the solid, particles vibrate and transfer energy, but do not travel down the strip.

Convection in water and air



Activity 7.3: A potassium-permanganate streak rises where water is heated and comes down along cooler regions, revealing circulation. In air, the paper cup above the candle rises because the warmed air expands and becomes less dense.

Sea breeze versus land breeze

	Sea breeze (day)	**Land breeze (night)**
Warmer surface	Land	Sea
Surface air flows	Sea → land	Land → sea
Reason	Warm air rises above hot land; cooler sea air replaces it.	Land cools faster; warmer air rises above sea.



Safety check

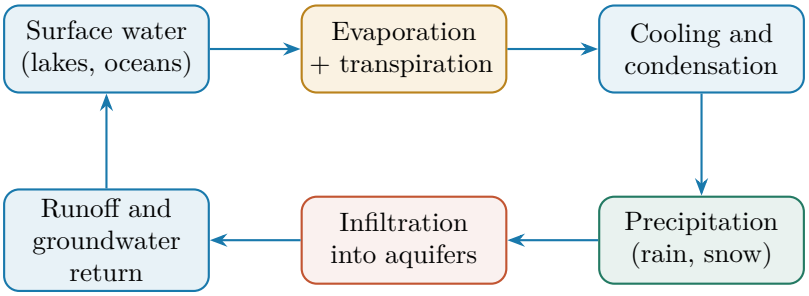
Do candle, metal-strip, boiling-water and potassium-permanganate activities **only with a teacher/adult**. Do not handle hot glass, heated metal or flames yourself.

Radiation, buildings and the water cycle

Radiation and everyday insulation

Heat from the Sun reaches Earth across space by **radiation**. This is why radiation is different from conduction and convection: it needs **no material medium**. Light-coloured surfaces can reflect more sunlight; trapped air in wool, double layers or hollow bricks reduces heat transfer.

Water cycle: follow the arrows



Two new words from the chapter

Infiltration: movement of surface water downward through soil and rock openings.
Aquifer: an underground layer of permeable sediment or rock that stores groundwater.
Conservation: recharge pits and rainwater harvesting can help replenish groundwater. In Ladakh, *ice stupas* store frozen water that melts during warmer months.

Project ideas (textbook p. 104)

Ready-to-try observations

Rainwater harvesting: report where roof runoff goes, how a filter/recharge pit works, and add your own sketch.
Paper on a metal rod: the metal spreads away heat; paper wrapped tightly around it may resist ignition briefly (adult demonstration only).
Paper spiral above a candle: rising warm air can rotate the spiral (teacher demonstration only). These activities illustrate conduction and convection.

One-minute revision and mistakes to avoid

Memory picture: C–C–R

Conduction: contact and neighbouring particles (mainly solid).
Convection: fluid circulates (liquid or gas).
Radiation: rays can pass through empty space.

Common confusions

- Do not say *heat rises*. **Hot fluids** tend to rise; conduction can transfer heat sideways or downwards.
- **Land breeze** is from *land to sea*, commonly at night; **sea breeze** is the opposite.
- **Sand** generally permits more infiltration than **clay**; do not interchange them.
- Do not say wood cannot transfer any heat. It is a **poor conductor**, not a perfect insulator.

Quick self-test (answers follow)

(1) Which heat-transfer mode does not need matter? (2) At night, which way does a land breeze move?
(3) What are underground water-storing layers called? (4) In the metal-strip activity, which pin falls first?

Answers: (1) Radiation. (2) Land to sea. (3) Aquifers. (4) Pin I, nearest the flame.

A little more practice: explain in your own words

Challenge 1 • Steel spoon or wooden stirrer?

You leave a steel spoon and a wooden stirrer in hot soup for one minute. Which handle is likely to feel hotter, and why?
Answer. The **steel spoon** generally feels hotter because it conducts thermal energy from the soup along its length more efficiently than wood does.

Challenge 2 • Why do we wear layered clothes?

Answer. Layers trap pockets of air. Air conducts heat poorly, reducing the rate at which thermal energy escapes from our bodies.

Challenge 3 • Clouds and rain

A puddle becomes smaller after sunshine, but later clouds form and rain falls. Name the three processes.
Answer. **Evaporation** at the puddle, **condensation** when vapour forms clouds, and **precipitation** when water falls as rain.

One final comparison to remember

Conduction transfers energy through contact; **convection** transfers energy by motion of fluids; **radiation** transports energy without needing a material medium.
When answering an NCERT reasoning question, name the *method* first and then explain *what moves and why.*